higher dimensional cousins in Figure 13.2, come with strings attached, literally, and hence their interactions are governed by strings. Not surprisingly, then, zero-branes behave very differently from conventional point particles, and, most important, they participate fully in the spreading out and lessening of ultramicroscopic spacetime jitters; zero-branes do not reintroduce the fatal flaws afflicting point-particle schemes that attempt to merge quantum mechanics and general relativity.

In fact, Tom Banks of Rutgers University and Willy Fischler of the University of Texas at Austin, together with Leonard Susskind and Stephen Shenker, both now at Stanford, have formulated a version of string/M-theory in which zero-branes are *the* fundamental ingredients that can be combined to generate strings and the other, higher dimensional branes. This proposal, known as *Matrix theory*—still another possible meaning for the "M" in "M-theory"—has generated an avalanche of follow-up research, but the difficult mathematics involved has so far prevented scientists from bringing the approach to completion. Nevertheless, the calculations that physicists have managed to carry out in this framework seem to support the proposal. If Matrix theory is true, it might mean that everything—strings, branes, and perhaps even space and time themselves—is composed of appropriate aggregates of zero-branes. It's an exciting prospect, and researchers are cautiously optimistic that progress over the next few years will shed much light on its validity.

We have so far surveyed the path string theorists have followed in the search for spacetime's ingredients, but as I mentioned, there is a second path coming from string theory's main competitor, loop quantum gravity. Loop quantum gravity dates from the mid-1980s and is another promising proposal for merging general relativity and quantum mechanics. I won't attempt a detailed description (if you're interested, take a look at Lee Smolin's excellent book *Three Roads to Quantum Gravity*), but will instead mention a few key points that are particularly illuminating for our current discussion.

String theory and loop quantum gravity both claim to have achieved the long-sought goal of providing a quantum theory of gravity, but they do so in very different ways. String theory grew out of the successful particle physics tradition that has for decades sought matter's elementary